NO.262680104504

中国认可
检验
INSPECTION
CNAS IB0071

SAFETYDATASHEET

Product Name:	Rechargeable Li-ion Battery Pack B05BAT_1 (22.2V 2200mAh 48.84Wh)
Effective Date:	2026-01-11
Compiler:	Chen Yushuang
Checker:	Liu Wanqing
Approver:	Dongxuesheng

Shanghai Institute of Chemical Industry Testing Co., Ltd.

声　明

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Suzhou Clean-well Appliance Co., Ltd

SAFETY DATA SHEET

Rechargeable Li-ion Battery Pack B05BAT_1（22.2V 2200mAh 48.84Wh）

SECTION1 IDENTIFICATION

Product name:	Rechargeable Li-ion Battery Pack B05BAT_1 (22.2V 2200mAh 48.84Wh)
Company:	Suzhou Clean-well Appliance Co., Ltd
Address:	No. 79, Tongxin Road , Suzhou New District , Suzhou City, Jiangsu, China
E-mail:	Max. shen@szcleanwell.com
Fax:	0512-68585751
Emergency Phone:	0512-68585751
Recommend use of the chemical and restrictions on use:	/
SDS Number:	262680104504
Effective Date:	2026-01-11

SECTION2 HAZARDS IDENTIFICATION

The product is outside of the scope of GHS system.

Main Hazards:

Fire or Explosion Hazards:

Lithium ion battery contains flammable liquid electrolyte that may vent, ignite and produce sparks when subjected to high temperatures (>150℃), when damaged or abused (e.g., mechanical damage or electrical overcharging). May burn rapidly with flare-burning effect. May ignite other batteries in close proximity.

Health Hazards:

Contact with battery's electrolyte may be irritating to skin, eyes and mucous membranes. Fire will produce irritating, corrosive and/or toxic gases. Fumes may cause dizziness or suffocation.

SECTION3 INFORMATION ON INGREDIENTS

Product name: Rechargeable Li-ion Battery Pack B05BAT_1 (22.2V 2200mAh 48.84Wh)

Ingredient	Concentration(%)	CAS NO.	EC NO.
Nickel cobalt manganese lithium oxide	41	182442-95-1	695-690-9
Graphite	22	7782-42-5	231-955-3
Electrolyte (Lithium hexafluorophosphate)	16	21324-40-3	244-334-7
Copper	11	7440-50-8	231-159-6
Aluminium	5	7429-90-5	231-072-3
Dissepiment(polyethylene)	1	9002-88-4	618-339-3
Binder(PVDF)	1	24937-79-9	607-458-6
Binder Carboxymethylcellulose sodium	1	9004-32-4	618-378-6
Binder(Styrene-butadiene rubber)	1	9003-55-8	685-145-3
Conductive agent(Carbon Black)	1	1333-86-4	215-609-9

SECTION4 FIRST-AID MEASURES

Skin Exposure:

If in contact with the internal materials of battery, remove the contaminated clothing, shoes and socks, immediately flush with plenty of water for at least 20 minutes. Call a physician.

Eye Exposure:

In case of the internal battery's materials in contact with eyes, lift your eyelids immediately and rinse them with running water for more than 20 minutes. If irritation persists, call a physician.

Inhalation Exposure:

If inhaled the internal battery's materials, immediately remove to fresh air. If breathing is difficult, give oxygen. If not breathing, give artificial respiration. Call a physician.

Oral Exposure:

If swallowed the internal battery's materials, do not induce vomiting. Call a physician immediately.

Most Important Symptoms/Effects, Acute and Delayed:

No data available.

Indication of Immediate Medical Attention and Special Treatment Needed, if Necessary:

No data available.

SECTION5 FIRE FIGHTING MEASURES

Suitable Extinguishing Media:

Suitable:Water spray or regular foam.

Specific Hazards Arising from the Chemical:

May decompose upon combustion to generate irritating, corrosive or toxic fumes. Fumes may cause dizziness or suffocation.

Special Protective Action for Fire-fighters:

Protective Equipment: Wear self-contained breathing apparatus and protective clothing to prevent contact with skin and eyes. Fire-extinguishing work is done from the windward. Uninvolved persons should evacuate to a safe place.

SECTION6 ACCIDENTAL RELEASE MEASURES

Personal Precautions, Protective Equipment and Emergency Procedures:

Use personal protective equipment. Ensure adequate ventilation. Keep people away from and upwind of spill/leak. Entry to noninvolved personnel should be controlled around the leakage area by roping off. Remove all sources of ignition.

Environmental Precautions:

Avoid leakage enter the earth, ditches or waters. Avoid directly release the cleaning waste water into the environment.

Methods and Materials for Containment and Cleaning up:

If the electrolyte leaks, use soil, sand or other non-combustible materials to absorb. The leaked batteries and dirty adsorbents should be placed in metal containers.

SECTION7 HANDLING AND STORAGE

Precautions for Safe Handling:

Operators should be trained and strictly abide by operating procedures. Wear appropriate protective clothing and safety gloves. Keep away from ignition sources, heat and flame. No smoking at working site. Handling is performed in a well ventilated place. Avoid disassembling the battery at will and reversing battery polarity within the battery assembly. Such batteries must be packed in inner packaging in such a manner as to effectively prevent short circuits and to prevent movement which could lead to short circuits. In case of leakage of the material in the battery, avoid directly contact with eyes and skin. Avoid inhalation. Incompatibilities: Strong oxidizing agents, Combustible materials and Corrosives.

Conditions for Safe Storage, Including Any Incompatibilities:

Store in a cool, dry, and well-ventilated area. Keep away from ignition sources, heat and flame. Incompatibilities: Strong oxidizing agents, Combustible materials and Corrosives. Such batteries must be packed in inner packaging in such a manner as to effectively prevent short circuits and to prevent movement which could lead to short circuits. Storage place should be equipped with appropriate varieties and quantities of fire fighting equipment and leakage emergency treatment equipment.

SECTION8 EXPOSURE CONTROL/PPE

Control Parameteeters :

GBZ 2.1-2019 Occupational Exposure Limits for Hazardous Agents in the Workplace - Part 1:

Nickel cobalt manganese lithium oxide:Cobalt and compounds, as Co PC-TWA 0.05mg/m³ PC-STEL 0.1mg/m³ Remarks G2B; Sensitization
Nickel cobalt manganese lithium oxide:Manganese and inorganic compounds, as MnO2 PC-TWA 0.15mg/m³
Nickel cobalt manganese lithium oxide:Soluble nickel compounds PC-TWA 0.5mg/m³ Remarks G1(Nickel compounds), Sensitization
Graphite:Graphite dust PC-TWA(Total dust) 4mg/m³ PC-TWA(Inhalable dust) 2mg/m³
Copper:Copper, as Cu Copper dust PC-TWA 1 mg/m³ Copper fume PC-TWA 0.2mg/m³
Aluminium:Aluminum dust: Metal & alloys dust PC-TWA(Total dust) 3mg/m³
Dissepiment(polyethylene):Polyethylene dust PC-TWA(Total dust) 5mg/m³
Conductive agent(Carbon Black):Carbon black dust PC-TWA(Total dust) 4mg/m³ Remarks G2B
ACGIH:
Graphite:Graphite, natural TLV-TWA 2 mg/m³, R
Copper:Copper dusts and mists TLV-TWA 1 mg/m³ Copper fume TLV-TWA 0.2mg/m³
Aluminium:Aluminum TLV-TWA 1 mg/m³, R
Conductive agent(Carbon Black):Carbon black TLV-TWA 3 mg/m³, I

Appropriate Engineering Controls:

Mechanical exhaust required. Safety shower and eye bath.

Individual Protection Measures:

Eye/Face Protection:

Wear chemical safety glasses if needed.

Skin Protection:

Hand Protection: Wear safety gloves.
Body Protection: Wear appropriate protective clothing.

Respiratory Protection:

Wear government approved respirator if needed.

Thermal Hazards:

No data available.

Other Protect:

No smoking, drinking and eating at working site. Wash thoroughly after handling.

SECTION9 PHYSICAL/CHEMICAL PROPERTIES

Appearance:	Gray black Plastics cement shell
Odor:	Odorless
pH Value:	8-9
Solubility:	partial soluble in water
Melting Point/Freezing Point:	＞300℃
Boiling Point, Initial Boiling Point and Boiling Range:	No data available.

Flash Point(Closed Cup):	No data available.
Density/Relative Density:	No data available.
Kinematic Viscosity:	No data available.
Lower/Upper Explosion Limit/Flammabili ty Limit:	No data available.
Vapour Pressure:	No data available.
Relative Vapor Density:	No data available.
Partition Coefficient N-Octanol/Water(Log Value):	No data available.
Autoingnition Temperature:	No data available.
Decomposition Temperature:	No data available.
Particle Characteristics:	No data available.
Flammability(Solid, Gas):	No data available.

SECTION10 STABILITY AND REACTIVITY

Reactivity:

No data available.

Chemical Stability:

Stable under normal temperatures and pressures.

Possibility of Hazardous Reactions:

No data available.

Conditions to Avoid:

Avoid misoperation, exposure to heat and open flame. Avoid mechanical or electrical abuse and overcharge. Prevent short circuits. Prevent movement which could lead to short circuits.

Incompatible Materials:

Strong oxidizing agents, Combustible materials and Corrosives.

Hazardous Decomposition Products:

Carbon oxides, metal oxides, etc.

SECTION11 TOXICOLOGICAL INFORMATIO

Acute Toxicity:

No data available.

Skin Corrosion/Irritation:

The electrolyte in the battery causes skin irritation.

Serious Eye Damage/Irritation:

The electrolyte in the battery causes eye irritation.

Respiratory Sensitization:

No data available.

Carcinogenicity:

No data available.

Skin Sensitization:

No data available.

Germ Cell Mutagenicity:

No data available.

Reproductive Toxicity:

No data available.

Specific Target Organ Toxicity -Single Exposure:

No data available.

Specific Target Organ Toxicity -Repeated Exposure:

No data available.

Aspiration Hazard:

No data available.

SECTION12 ECOLOGICAL INFORMATION

Toxicity:

No data available.

Persistence and Degradability:

No data available.

Bioaccumulative Potential:

No data available.

Mobility in Soil:

No data available.

Other Adverse Effects:

No data available.

SECTION13 DISPOSAL CONSIDERATION

Disposal Methods:

The disposal of discarded battery shall comply with the requirements of relevant laws, regulations, policies and standards such as the "Law of the People's Republic of China on the Prevention and Control of Environmental Pollution by Solid Waste" and "Technical Policy for the Prevention and Control of Waste Battery Pollution". Contact a licensed professional waste disposal service to dispose of wastes. Used battery being transported for disposal or reclamation should be carefully checked prior to shipment to ensure the integrity of each battery and its

suitability for transport.

SECTION14 TRANSPORT INFORMATION

Lithium Battery Packed with Vacuum CLeaner during Transport:	The product has passed the test items of Manual of Test and Criteria Section 38.3 and UN Model Regulations, SP188, 1.2m drop test. The total net weight of the lithium batteries is less than 5 kg.
RID/ADR(2025 Edition):	The product is not subject to the other provisions of RID/ADR according to special provision 188. According to 2.2.9.1.7(g), Manufacturers and subsequent distributors of cells or batteries manufactured shall make available the test summary as specified in the Manual of Tests and Criteria, Part III, sub-section 38.3, paragraph 38.3.5.
IATA DGR(67th Edition):	The product shall meet the General Requirements and section II of Packaging Instruction 966. The package must be capable of withstanding the stacking test required in PI 966 II. According to 3.9.2.6.1(g), Manufacturers and subsequent distributors of cells or batteries manufactured after 30 June 2003 shall make available the test summary as specified in the Manual of Tests and Criteria, Part III, sub-section 38.3, paragraph 38.3.5.
IMO IMDG CODE(2024 Edition):	The product is not subject to the other provisions of IMO IMDG CODE according to special provision 188. According to 2.9.4.7, Manufacturers and subsequent distributors of cells or batteries manufactured shall make available the test summary as specified in the Manual of Tests and Criteria, Part III, sub-section 38.3, paragraph 38.3.5.

SECTION15 REGULATORY INFORMATION

Domestic Regulations:

Lithium Battery Packed with Equipment during Transport:

The product has passed the test items of Manual of Test and Criteria Section 38.3 and UN Model Regulations, SP188, 1.2m drop test. The total net weight of the lithium batteries is less than 5 kg.

Regulations Concerning Road Transportation of Dangerous Goods(JT/T 617-2018):

UN Number: UN3481

Name and Description: Lithiumion batteries packed with equipment

The product is not subject to JT/T 617-2018 according to special provision 188.

List of Dangerous Goods(GB 12268-2025):

UN Number: UN3481

Shipping Name: Lithium ion batteries packed with equipment

The product is not subject to GB 12268-2025 according to special provision 188.

List of Dangerous Goods by Rail(TB/T 30006-2022):

Number: 91046

Name of product: Lithium ion batteries packed with equipment

The product is not subject to TB/T 30006-2022 according to special provision 78, 79.

International Regulations:

Directive (EU)2023/1542 and 2013/56/EU:

The label, disposal and recycling of the battery shall meet the requirements of EU Directive (EU)2023/1542 and 2013/56/EU.

ICAO TI:

1. Unless be exempted according to ICAO TI, the lithium ion cell/batteries (UN 3480, PI 965) and lithium metal cell/batteries (UN 3090, PI 968) are forbidden for carriage on passenger aircraft.

2. Unless be approved according to ICAO TI, Lithium ion cells/batteries (UN 3480, PI 965) must be offered for transport at a state of charge (SoC) not exceeding 30% of their rated design capacity.

SECTION16 OTHER INFORMATION

Preparation Date:

2026-01-11

Preparation Department:

Shanghai Institute of Chemical Industry Testing Co., Ltd.
Tel(Fax):+86-21-52815377/31765555

Revision:

0

Abbreviations and Acronyms:

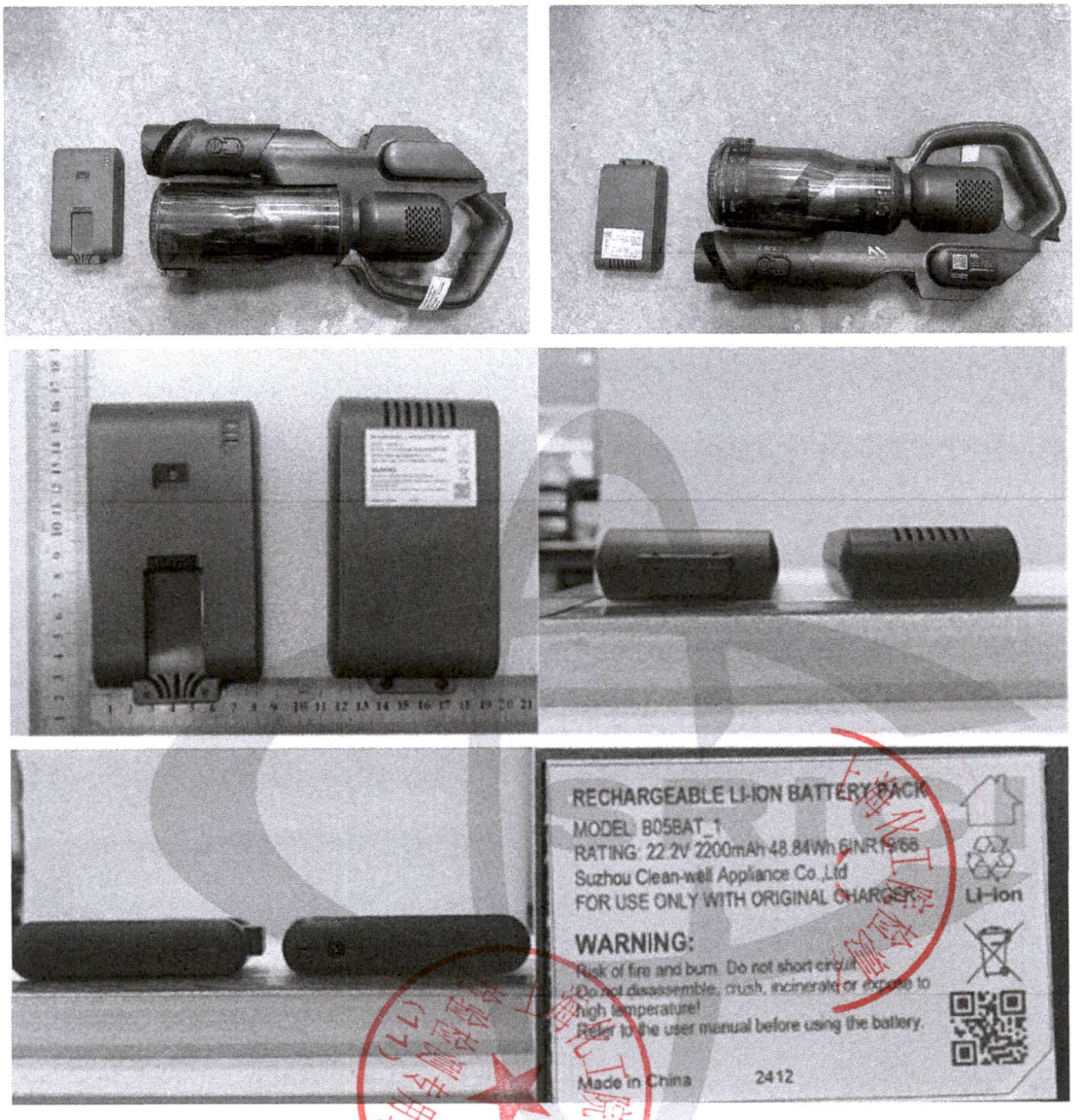
CAS: Chemical Abstracts Service EC: European Commission ACGIH: American Conference of Governmental Industrial Hygienists PC-TWA: Permissible concentration-time weighted average PC-STEL: Permissible concentration-short term exposure limit G1: Carcinogenic to humans G2B: Possibly carcinogenic to humans Sensitization: The substance may have allergenic effects TLV-TWA: Time weighted average threshold limit R: Measured as respirable fraction of the aerosol RID: Regulations concerning the International Carriage of Dangerous Goods by Rail ADR: European Agreement concerning the International Carriage of Dangerous Goods by Road IMO IMDG CODE: International Maritime Organization International Maritime Code for Dangerous Goods IATA DGR: International Air Transport Association Dangerous Goods Regulations EU: European Union ICAO TI: International Civil Aviation Organization Technical Instructions for the Safe Transport of Dangerous Goods by Air PI: Packaging Instruction I: Measured as Inhalable fraction of the aerosol

Other Information:

This SDS is only compiled for battery and based on the information such as ingredients provided by the applicant and our current knowledge. This SDS shall be used only as a guide. If the battery is used as a component in another product, the information in this SDS may not be applicable. The users of this SDS must make independent judgments on the correctness and completeness and then

decide its suitability according to the actual situation. The users should take the relevant legal responsibilities for the consequences of use.

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END OF REPORT